CNN/Siamese Embedding Endpoints

Integration guide for the Flutter team — a second, independent matching path (single-finger and slap) alongside the existing minutiae endpoints

PLEASE READ: the slap embedding threshold was just recalibrated (0.94 -> 0.78, real-photo-based, see §7). When you test slap again, please report specifically **which way it fails, if it still does** — rejecting people who should pass (too strict) or letting the wrong person through (too loose). That direction matters more than "it still doesn't work" — it tells us which way to adjust next.

1. What this is

Nine new endpoints (three single-finger, three slap-embedding, three slap-hybrid) on the same cloud service you already talk to, offering alternatives to minutiae-based matching: a CNN (pretrained ResNet18) converts each fingerprint image into a 512-number vector ("embedding"), and two images are compared by how similar their vectors are (cosine similarity), instead of comparing extracted ridge-point locations. The single-finger version is validated at 92.5% matching accuracy on a 336-person test set (vs. ~52% for the existing minutiae matcher) — see the attached *Siamese_Embedding_Phase_A_Report.pdf* for the full methodology and numbers. The slap-embedding version runs that same single-finger comparison once per finger (see §4); slap-hybrid additionally combines it with the minutiae matcher per finger (see §5). Neither slap version has been separately measured for slap-specific accuracy/FAR/FRR on real multi-finger captures — treat both as functionally verified, not yet accuracy-validated at the slap level.
This runs **alongside** the existing endpoints, not instead of them. Nothing about /v2/enroll, /v2/authenticate, /v2/verify, /v2/enroll_slap, /v2/authenticate_slap, or /v2/verify_slap changes — they still work exactly as before, on their own separate data tables.

2. Base URL

`https://34-100-150-103.sslip.io`
Same host as the existing /v2/* minutiae endpoints.

3. Single-finger endpoints

POST /v2/enroll_embedding

Multipart form fields:

Field	Required	Notes
image	Yes	The preprocessed fingerprint image (same as you'd send to /v2/enroll)
name	Yes	Person's name
uid	Yes	Unique person identifier
batch	Yes	Batch/group identifier — authenticate matches within one batch
finger_position	No	Integer 0-10, default 0

Response (success):

```
{"success": true, "embedding_dim": 512}
```

Response (error, e.g. missing field or bad image):

```
{"success": false, "error": "..."}
```

POST /v2/authenticate_embedding — 1:N (identify)

Multipart fields: **batch**, **image**. Compares the image against every person enrolled in that batch, returns the best match.

Response (match found):

```
{"success": true, "name": "...", "uid": "...", "confidence": 0.9954, "threshold": 0.94}
```

Response (no match / below threshold):

```
{"success": false, "message": "No match found", "confidence": 0.6123, "threshold": 0.94}
```

POST /v2/verify_embedding — 1:1 (confirm identity)

Multipart fields: **uid**, **batch**, **image**. Compares the image only against that specific person's enrolled embedding.

Response:

```
{"success": true, "matched": true, "confidence": 0.9954, "threshold": 0.94, "name": "...",
"uid": "..."}
```

Response (uid not found in batch):

```
{"success": false, "error": "No user found: <uid>"} — HTTP 404
```

4. Slap endpoints

Same request shape as the existing minutiae slap endpoints: multiple images in the image field (one per finger, in capture order), plus **hand_side** ("left"/"right", default "right") and an optional **finger_order** (comma-separated finger-position names) if your capture order doesn't match the default 4-finger layout. **Matching logic: per-finger, not averaged.** Each probe finger is compared only to its own same-position enrolled finger, and judged independently against its own threshold — no averaging across fingers. A candidate is accepted only if **at least 2 of their fingers individually clear it** (min_fingers_required below, server-configurable). A single strong finger score can no longer carry a weak one. **Note the threshold here is 0.78, not single-finger's 0.94** — recalibrated specifically for real-photo conditions (see §7); it's a different, independently-configurable number (EMBEDDING_SLAP_FINGER_THRESHOLD).

POST /v2/enroll_slap_embedding

Multipart form fields:

Field	Required	Notes
image	Yes (multiple)	One image per finger, same field name repeated
uid	Yes	Unique person identifier
name	No	Person's name
batch	No	Default "default"
hand_side	No	"left" or "right", default "right"
finger_order	No	Comma-separated finger-position override

Response:

```
{"success": true, "uid": "...", "name": "...", "batch": "...", "hand_side": "right",
"finger_count": 4, "enrolled_fingers": [{"finger_position": "right_little", "iso_code": 5,
"embedding_dim": 512}, ...]}
```

POST /v2/authenticate_slap_embedding — 1:N (identify)

Multipart fields: **batch**, **hand_side** (optional), **image** (multiple). Compares against every person enrolled in the batch.
"success" here means a candidate cleared min_fingers_required and is being returned — check fingers_passed/min_fingers_required for the actual decision detail, not a confidence-vs-threshold comparison.

Response (match found):

```
{"success": true, "uid": "...", "name": "...", "hand_side": "right", "fingers_passed": 2,
"min_fingers_required": 2, "avg_confidence": 0.9957, "threshold": 0.78, "matched_fingers":
[{"probe_position": "right_little", "matched_position": "right_little", "confidence":
0.9954, "passed": true}, ...]}
```

Response (no match — e.g. only 1 of 2 fingers passed):

```
{"success": false, "message": "No match found", "fingers_passed": 1,
"min_fingers_required": 2, "threshold": 0.78, "matched_fingers": [...]}
```

POST /v2/verify_slap_embedding — 1:1 (confirm identity)

Multipart fields: **uid**, **batch**, **hand_side** (optional), **image** (multiple). Compares only against that specific person's enrolled fingers. Use **matched** for the pass/fail decision — it's already fingers_passed >= min_fingers_required, no need to recompute it client-side.

Response:


```
{"success": true, "matched": true, "uid": "...", "name": "...", "hand_side": "right",
"fingers_passed": 2, "min_fingers_required": 2, "avg_confidence": 0.9957, "threshold":
0.78, "matched_fingers": [{"probe_position": "right_little", "matched_position":
"right_little", "confidence": 0.9954, "passed": true}, {"probe_position": "right_ring",
"matched_position": "right_ring", "confidence": 0.9881, "passed": true}]}
```

5. Slap-hybrid endpoints (minutiae + embedding fused)

Same request shape as the slap-embedding endpoints above (multiple images, one per finger, **hand_side**/**finger_order**). The difference is what happens per finger: this path runs **both** the old minutiae matcher and the new embedding matcher on every finger, combines the two scores into one **fused score**, and judges that fused score against its own threshold. Enrollment is slower than the other two paths (it runs both extractors), but request/response shape is otherwise the same family.

Why fuse them at all? We checked whether minutiae and embedding actually make different mistakes on the same real photos, or just fail on the same ones. They fail on partly different ones — combining them (properly weighted: 80% embedding, 20% minutiae, fit by measuring real data, not guessed) measured 85.37% accuracy vs. 82.73% for embedding alone on the same 30 real people — a real but modest gain, not a dramatic one. Same acceptance rule as slap-embedding: a candidate passes if at least 2 fingers individually clear the fused threshold.

POST /v2/enroll_slap_hybrid

Multipart form fields: same as /v2/enroll_slap_embedding.

Response:

```
{"success": true, "uid": "...", "name": "...", "batch": "...", "hand_side": "right",
"finger_count": 2, "enrolled_fingers": [{"finger_position": "right_little", "iso_code": 5,
"minutiae_count": 83, "embedding_dim": 512}, ...]}
```

POST /v2/authenticate_slap_hybrid — 1:N (identify)

Multipart fields: same as /v2/authenticate_slap_embedding. Note the field is **fused_threshold**, not **threshold** — the fused score lives on a different scale (roughly -3 to +4, not 0-1) since it's a weighted combination of two z-normalized scores, not a raw similarity.

Response (match found):

```
{"success": true, "uid": "...", "name": "...", "hand_side": "right", "fingers_passed": 2,
"min_fingers_required": 2, "avg_fused_score": 2.68679, "fused_threshold": 1.1162,
"matched_fingers": [{"probe_position": "right_little", "matched_position": "right_little",
"fused_score": 2.78227, "minutiae_score": 0.12493, "embedding_score": 0.92298, "passed":
true}, ...]}
```

Response (no match):

```
{"success": false, "message": "No match found", "fingers_passed": 1,
"min_fingers_required": 2, "fused_threshold": 1.1162, "matched_fingers": [...]}
```

POST /v2/verify_slap_hybrid — 1:1 (confirm identity)

Multipart fields: same as /v2/verify_slap_embedding. Use **matched** for the pass/fail decision.

Response:


```
{"success": true, "matched": true, "uid": "...", "name": "...", "hand_side": "right",
"fingers_passed": 2, "min_fingers_required": 2, "avg_fused_score": 2.68679,
"fused_threshold": 1.1162, "matched_fingers": [{"probe_position": "right_little",
"matched_position": "right_little", "fused_score": 2.78227, "minutiae_score": 0.12493,
"embedding_score": 0.92298, "passed": true}, {"probe_position": "right_ring",
"matched_position": "right_ring", "fused_score": 2.59131, "minutiae_score": 0.05141,
"embedding_score": 0.93943, "passed": true}]}
```


A concrete example of the fusion actually helping, from our own testing: two impostor fingers scored 0.753 and 0.796 on embedding alone — both uncomfortably close to slap-embedding's 0.78 threshold, borderline false-accept risk. Minutiae's scores for those same two fingers were 0.022 and 0.0 — clearly "different finger." The fused score pulled both well below the fused threshold, turning a near-miss into a clear, comfortable rejection.

6. Reading the response

confidence (and each finger's individual score, in matched_fingers) is a cosine similarity score from -1 to 1 (in practice you'll see roughly 0.85-1.0 for fingerprint photos, genuine or not — this embedding space sits in a narrow, high band, not spread across the full range).
Single-finger decision: matched is a direct confidence-vs-threshold comparison (currently 0.94, returned in every response so the app never needs to hardcode it).
Slap decision works differently — it's a count, not a magnitude comparison. Each finger is checked against threshold independently (matched_fingers[i].passed), and the response is only a match when fingers_passed >= min_fingers_required (default 2). Don't try to compare the slap-level "confidence"/"avg_confidence" field against threshold yourself to derive a decision — it's informational (useful for showing the user a number), the actual decision is already made and returned for you.

7. Important — nothing here is fully calibrated yet

Single-finger's 0.94 was measured on a lab dataset (HKPU contactless fingerprint database — different camera, lighting, and preprocessing than this app's actual capture pipeline). It's a reasonable starting point, not a final production number — build against the **threshold** field in each response rather than hardcoding 0.94 anywhere in the app, so a server-side recalibration doesn't require a client update.

Slap-embedding's threshold (0.78) was recalibrated specifically from real, uncontrolled photos rather than the lab dataset — a better-grounded starting point than reusing 0.94, but still not measured against actual multi-finger slap captures. **This is why we need your test results** (see the box at the top of this document): please report which direction it fails, if it still does.

min_fingers_required = 2 (env-configurable, EMBEDDING_SLAP_MIN_FINGERS / HYBRID_SLAP_MIN_FINGERS) is a reasonable-sounding rule for both slap paths — require more than one finger to individually pass — but hasn't been measured against real slap captures either. Whether 2-of-4 is the right cutoff (vs. 1-of-4, 3-of-4) is open. Build against **min_fingers_required** in the response — don't hardcode "2" anywhere in the app.

The slap-hybrid fusion weights (80% embedding / 20% minutiae) and fused threshold (1.1162) were fit on the same 30-person real-photo dataset as the slap-embedding threshold — real data, not a guess, but still a single small dataset of individual photos, not actual multi-finger slap captures. Treat it the same way: a well-reasoned starting point, not a finished number.

8. Example (curl)

Enroll (single-finger):

```
curl -X POST https://34-100-150-103.sslip.io/v2/enroll_embedding \
  -F "name=Jane Doe" -F "uid=U001" -F "batch=demo" \
  -F "image=@fingerprint.jpg"
```

Verify (single-finger):

```
curl -X POST https://34-100-150-103.sslip.io/v2/verify_embedding \
  -F "uid=U001" -F "batch=demo" -F "image=@fingerprint2.jpg"
```

Enroll (slap-embedding, 4 fingers):

```
curl -X POST https://34-100-150-103.sslip.io/v2/enroll_slap_embedding \
  -F "uid=U001" -F "name=Jane Doe" -F "batch=demo" -F "hand_side=right" \
  -F "image=@little.jpg" -F "image=@ring.jpg" \
  -F "image=@middle.jpg" -F "image=@index.jpg"
```

Enroll (slap-hybrid, same fields, different endpoint):

```
curl -X POST https://34-100-150-103.sslip.io/v2/enroll_slap_hybrid \
  -F "uid=U001" -F "name=Jane Doe" -F "batch=demo" -F "hand_side=right" \
  -F "image=@little.jpg" -F "image=@ring.jpg" \
  -F "image=@middle.jpg" -F "image=@index.jpg"
```

9. Which one should the app actually call?

Three independent matchers are live now for slap: minutiae-only (/v2/*_slap), embedding-only (/v2/*_slap_embedding), and hybrid (/v2/*_slap_hybrid). None share enrollment data — enrolling in one doesn't enroll in the others, so a person needs separate enrollment calls if you want to compare all three. If you want to compare them side-by-side during evaluation, call all relevant endpoints with the same image(s) and show the results together; for a single production flow, use whichever one the team decides on after real usage data comes in (see §7) — that decision isn't made yet, so don't hardcode a permanent choice. Hybrid measured better than embedding-alone on our test data (85.37% vs. 82.73%), but the gap is modest, not dramatic, and neither has been validated on real slap captures — don't treat hybrid as automatically "the good one" without your own testing confirming it.

Backend: backend/embedding_matcher.py + backend/slap_embedding_matching.py + backend/hybrid_matcher.py + backend/slap_hybrid_matching.py + backend/cloud_app.py (branch phone-spoof). Validation report: siamese_embedding/Siamese_Embedding_Phase_A_Report.pdf. Hybrid calibration: siamese_embedding/compare_matchers_30.py, fusion_params.json.